

TUNKU ABDUL RAHMAN UNIVERSITY OF MANAGEMENT AND TECHNOLOGY

FACULTY OF COMPUTING AND INFORMATION TECHNOLOGY

ACADEMIC YEAR 2024/2025

OCTOBER EXAMINATION

AMIS1012 ETHICS IN COMPUTING

SATURDAY, 19 OCTOBER 2024

TIME: 9.00 AM – 11.00 AM (2 HOURS)

DIPLOMA IN COMPUTER SCIENCE

DIPLOMA IN INFORMATION TECHNOLOGY

Instructions to Candidates:

Answer **ALL** questions. All questions carry equal marks.

AMIS1012 ETHICS IN COMPUTING**Question 1**

- a) Discuss the differences among morals, ethics, and law. Provide an example for each term discussed. (9 marks)
- b) “Ethical decision-making is how people or groups choose the morally right action in a complex situation.”
Describe the **FIVE (5)** major steps involved in ethical decision-making. (5 marks)
- c) Define “exploits” in the context of a computer attack on a commercial system. Explain how a software developer responds to exploits. (4 marks)
- d) Discuss the following active attacks:
- (i) trojan horse (2 marks)
 - (ii) distributed denial of service attack (2 marks)
 - (iii) rootkit (2 marks)
- e) Identify the main objective of an attacker performs a passive attack on a company. (1 mark)

[Total: 25 marks]

Question 2

TAR Sdn. Bhd. is a retail business that sells home products, including kitchen gadgets, storage containers, and drinkware. The company has recently upgraded its point-of-sale (POS) system to a cloud-based solution. During the upgrading process, the business owner instructed the IT team to conduct a security risk assessment, emphasising the need to ensure the security of customer payment data.

- a) Examine why security risk assessment is important for TAR Sdn. Bhd. (3 marks)
- b) Outline the **EIGHT (8)** steps in a general security risk evaluation process that should be conducted by the IT team from TAR Sdn. Bhd. (8 marks)
- c) Based on the case above, identify the main asset that appears important to TAR Sdn. Bhd. (2 marks)

AMIS1012 ETHICS IN COMPUTING**Question 2 (Continued)**

- d) CIA triad principles are essential to be adopted in the new upgraded system. Discuss the **THREE (3)** CIA triad principles and provide a scenario for each principle to demonstrate how it should be applied in TAR Sdn. Bhd. (12 marks)

[Total: 25 marks]

Question 3

“Identifying and eliminating a defect early in the software development process will cost up to 100 times less than removing a defect in software that has already been shipped to consumers.”

- a) Do you agree with the statement above? Provide relevant supporting justifications to elaborate on your point of view. (5 marks)
- b) Examine how black-box testing and white-box testing can be used to eliminate defects in software. For each testing method, provide relevant examples to support your answer. (8 marks)
- c) With the aid of a diagram, discuss how an organisation applies the PDCA cycle to verify the quality of software goods or services. Each process in the cycle should be clearly explained. (10 marks)
- d) Describe a characteristic of good software. (2 marks)

[Total: 25 marks]

Question 4

Domitri is a renowned software developer from Malaysia. He developed an innovative algorithm for real-time data encryption, and software using this algorithm is widely adopted among technology companies. The algorithm is his proprietary creation, and he holds the rights to it. Domitri can decide who can utilise his algorithm; for example, if a software company wants to integrate his encryption technology into their products, he has the authority to permit or restrict its usage. Domitri can also legally protect his algorithm if someone attempts to use it without his permission.

- a) Examine why Intellectual Property Rights (IPR) are important to Domitri. Provide relevant justifications in your answer. (5 marks)
- b) Suggest to Domitri the **TWO (2)** most suitable types of IPR he should consider for his encryption algorithm. Provide relevant justification for your suggestion. (10 marks)

AMIS1012 ETHICS IN COMPUTING**Question 4 (Continued)**

- c) Domitri would like to register a trademark for his brand name. Explain to Domitri the **FIVE (5)** spectrums of trademarks. Provide a relevant example for each spectrum. (10 marks)

[Total: 25 marks]